

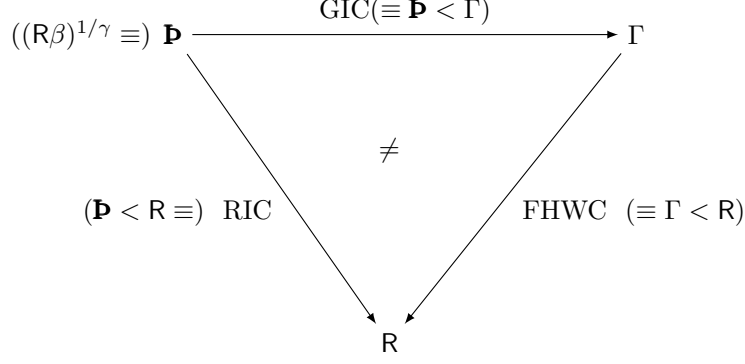


Figure 1 Appendix: Inequality Conditions for Perfect Foresight Model
(Start at a node and follow arrows)

1 Relational Diagrams for the Inequality Conditions

This appendix explains in detail the paper’s ‘inequalities’ diagrams (Figures 7, 8).

1.1 The Unconstrained Perfect Foresight Model

A simple illustration is presented in Figure 1, whose three nodes represent values of the absolute patience factor $\mathbf{P}$, the permanent-income growth factor $\mathcal{G}$, and the riskfree interest factor R . The arrows represent imposition of the labeled inequality condition (like, the uppermost arrow, pointing from $\mathbf{P}$ to $\mathcal{G}$, reflects imposition of the GIC condition (clicking GIC should take you to its definition; definitions of other conditions are also linked below)).¹ Annotations inside parenthetical expressions containing $\equiv$ are there to make the diagram readable for someone who may not immediately remember terms and definitions from the main text. (Such a reader might also want to be reminded that R, β , and Γ are all in $\mathbb{R}_{++}$, and that $\gamma > 1$).

Navigation of the diagram is simple: Start at any node, and deduce a chain of inequalities by following any arrow that exits that node, and any arrows that exit from successive nodes. Traversal must stop upon arrival at a node with no exiting arrows. So, for example, we can start at the $\mathbf{P}$ node and impose the GIC and then the FHCW, and see that imposition of these conditions allows us to conclude that $\mathbf{P} < R$.

One could also impose $\mathbf{P} < R$ directly (without imposing GIC and FHCW) by following the downward-sloping diagonal arrow exiting $\mathbf{P}$. Although alternate routes from one node to another all justify the same core conclusion ($\mathbf{P} < R$, in this case), $\neq$ symbol in the center is meant to convey that these routes are not identical in other respects. This

¹For convenience, the equivalent ($\equiv$) mathematical statement of each condition is expressed nearby in parentheses.

notational convention is used in category theory diagrams,² to indicate that the diagram is not commutative.³

Negation of a condition is indicated by the reversal of the corresponding arrow. For example, negation of the RIC, $\neg \text{RIC} \equiv \mathbf{D} > R$, would be represented by moving the arrowhead from the bottom right to the top left of the line segment connecting $\mathbf{D}$ and R .

If we were to start at R and then impose $\neg \text{FWC}$, that would reverse the arrow connecting R and $\mathcal{G}$, but the $\mathcal{G}$ node would then have no exiting arrows so no further deductions could be made. However, if we *also* reversed GIC (that is, if we imposed $\neg \text{GIC}$), that would take us to the $\mathbf{D}$ node, and we could deduce $R > \mathbf{D}$. However, we would have to stop traversing the diagram at this point, because the arrow exiting from the $\mathbf{D}$ node points back to our starting point, which (if valid) would lead us to the conclusion that $R > R$. Thus, the reversal of the two earlier conditions (imposition of $\neg \text{FWC}$ and $\neg \text{GIC}$) requires us also to reverse the final condition, giving us $\neg \text{RIC}$.⁴

Under these conventions, Figure 7 in the main text presents a modified version of the diagram extended to incorporate the PFFVAC.

This diagram can be interpreted, for example, as saying that, starting at the $\mathbf{D}$ node, it is possible to derive the PFFVAC⁵ by imposing both the GIC and the FHC; or by imposing RIC and $\neg \text{FWC}$. Or, starting at the $\mathcal{G}$ node, we can follow the imposition of the FHC (twice — reversing the arrow labeled $\neg \text{FWC}$) and then $\neg \text{RIC}$ to reach the conclusion that $\mathbf{D} < \mathcal{G}$. Algebraically,

$$\begin{aligned} \text{FWC} : \quad & \mathcal{G} < R \\ \neg \text{RIC} : \quad & R < \mathbf{D} \\ & \mathcal{G} < \mathbf{D} \end{aligned} \tag{1}$$

which leads to the negation of both of the conditions leading into $\mathbf{D}$. $\neg \text{GIC}$ is obtained directly as the last line in (1) and PFFVAC follows if we start by multiplying the Return Patience Factor ($\text{RPF} = \mathbf{D}/R$) by the FHW ($= \mathcal{G}/R$) raised to the power $1/\gamma - 1$, which is negative since we imposed $\gamma > 1$. FHC implies $\text{FWF} < 1$ so when FWF is raised to a negative power the result is greater than one. Multiplying the RPF (which exceeds 1 because $\neg \text{RIC}$) by another number greater than one yields a product that must

²For a popular introduction to category theory, see Riehl (2017).

³But the rest of our notation does not necessarily abide by the other conventions of category theory diagrams.

⁴The corresponding algebra is

$$\begin{aligned} \neg \text{FWC} : \quad & R < \mathcal{G} \\ \neg \text{GIC} : \quad & \mathcal{G} < \mathbf{D} \\ \Rightarrow \neg \text{RIC} : \quad & R < \mathbf{D}, \end{aligned}$$

⁵in the form $\mathbf{D} < (R/\mathcal{G})^{1/\gamma} \mathcal{G}$

be greater than one:

$$\begin{aligned}
1 &< \overbrace{\left(\frac{(R\beta)^{1/\gamma}}{R} \right)}^{>1 \text{ from RIC}} \overbrace{\left(\mathcal{G}/R \right)^{1/\gamma-1}}^{>1 \text{ from FHC}} \\
1 &< \left(\frac{(R\beta)^{1/\gamma}}{(R/\mathcal{G})^{1/\gamma} R \mathcal{G}/R} \right) \\
R^{1/\gamma} \mathcal{G}^{1-1/\gamma} &= (R/\mathcal{G})^{1/\gamma} \mathcal{G} < \mathbf{P}
\end{aligned}$$

which is one way of writing ~~PFFVAC~~.

The complexity of this algebraic calculation illustrates the usefulness of the diagram, in which one merely needs to follow arrows to reach the same result.

After the warmup of constructing these conditions for the perfect foresight case, we can represent the relationships between all the conditions in both the perfect foresight case and the case with uncertainty as shown in Figure 8 in the paper (reproduced here).

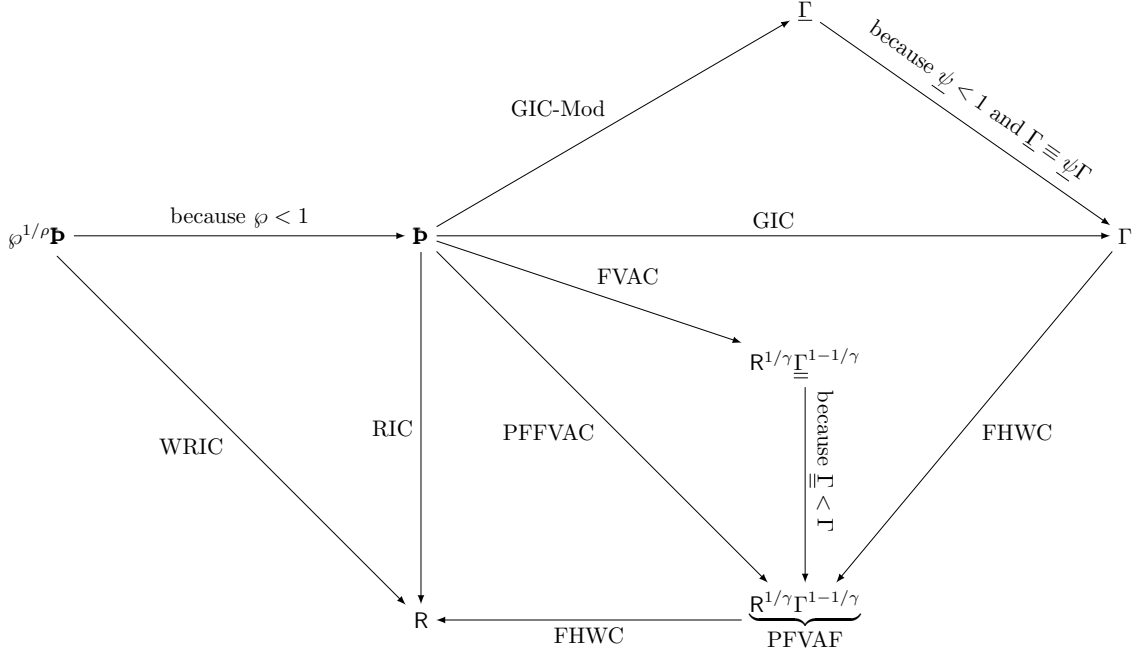


Figure 2 Appendix: Relation of All Inequality Conditions

Finally, the next diagram substitutes the values of the various objects in the diagram under the baseline parameter values and verifies that all of the asserted inequality conditions hold true.

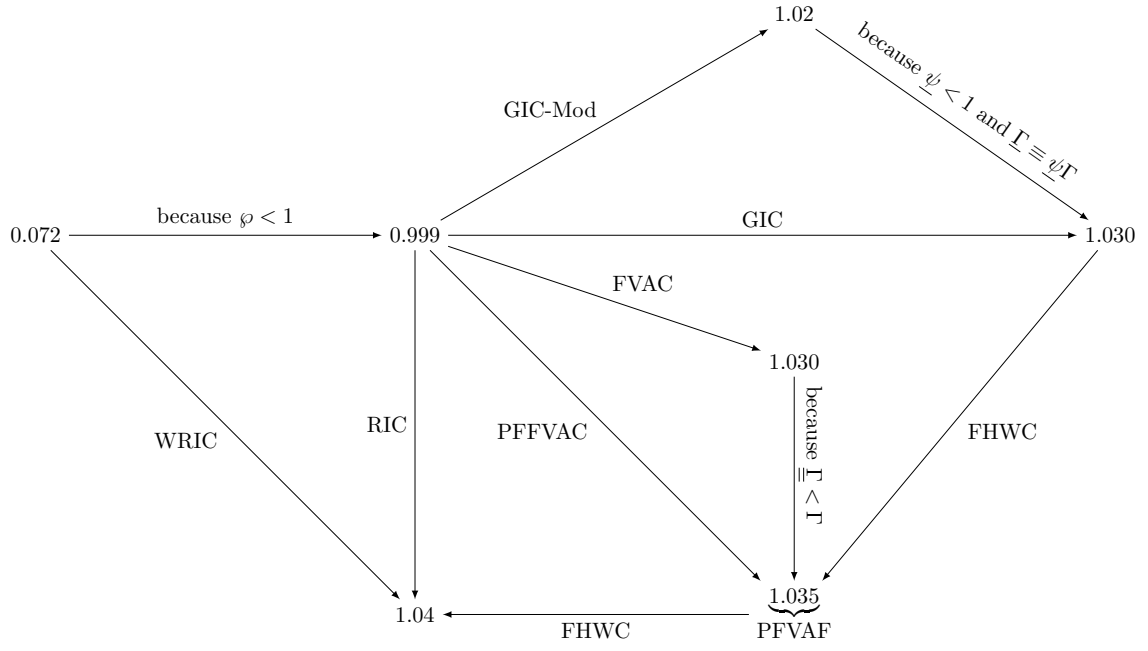


Figure 3 Appendix: Numerical Relation of All Inequality Conditions

References

RIEHL, EMILY (2017): Category theory in context. Courier Dover Publications.